# Charge trapping at MoS2-SiO2 interfaces

"How do MoS2 and SiO2 interface states trap charges in TENGs?" *Generated by LeapSpace*

**$MoS_2$ and $SiO_2$ interface states trap charge in TENGs by acting as localized energy wells that capture electrons or holes, hold them at the interface, and slow their release; in $MoS_2$/$SiO_2$-based TENGs, the strongest evidence points to $SiO_2$ dangling-bond states and $MoS_2$/dielectric interface defects as the key traps that create hysteresis, stabilize stored charge, and boost retained triboelectric output.** [59] [94] [95] [96] [103]

## How the trapping works

| Interface-state role | What it does | TENG consequence |
|---|---|---|
| **$SiO_2$ dangling bonds / surface defects** | Create very high-density trap sites and localized impurity levels inside the $MoS_2$ bandgap [59] | Charges injected by triboelectrification can be confined at the $MoS_2$/$SiO_2$ interface instead of rapidly dissipating [59] [103] |
| **$MoS_2$–$SiO_2$ defect states** | Single Si-dangling bonds act as **electron traps** and O-dangling bonds as **hole traps** [94] | Both polarities can be immobilized, producing hysteresis and threshold shifts that reflect stored interfacial charge [94] [95] |
| **Energy barriers / blocked traps** | Trap occupancy changes with temperature and barrier energies, so charges can become “blocked” or released [95] | Charge retention becomes time- and temperature-dependent, which directly affects TENG stability and output [95] |
| **Localized interfacial states** | Create charge confinement at the $MoS_2$/$SiO_2$ boundary and support charge-detrapping cycles [96] | Stored charge can be reused across sweeps or contact cycles, which is the basis for memory-like behavior in tribo/2D devices [96] |

## What this means in TENGs

- In triboelectric operation, contact electrification puts charge onto the surface; **interface states prevent that charge from immediately recombining or leaking away** by holding it in localized traps. [78] [97] [103]
- For **$MoS_2$**, trap states and localized defects are repeatedly described as charge-capture sites, and in $MoS_2$-based TENGs they are explicitly linked to **greater trapped-electron accumulation** and higher output. [98] [99] [100]

- For **$SiO_2$**, surface-state models and oxide trap literature show that interface and near-interface traps are stable charge reservoirs, which is why $SiO_2$ is a useful insulating/trapping layer in these devices. [59] [101] [102]
- In the combined **$MoS_2/SiO_2$** interface, the two materials work together: **$SiO_2$ provides dense trapping states**, while **$MoS_2$ supplies a 2D channel/sheet where trapped charge strongly perturbs transport**, producing hysteresis, threshold shifts, and strong electrostatic retention. [59] [103] [103]

## Bottom line

The interface traps in **$MoS_2$–$SiO_2$ TENGs** act like **charge reservoirs**: triboelectric charges are captured by dangling bonds and defect-derived localized states, then held long enough to enhance retained surface charge and output, but if the trap landscape becomes too conductive or too shallow, charges leak away faster and the benefit is reduced. [78] [94] [95] [103]

Confidence level: **High** — the retrieved sources consistently show charge trapping at the $MoS_2/SiO_2$ interface, identify specific defect types as electron/hole traps, and connect those traps to hysteresis, retained charge, and TENG performance.

## References


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